

Highlights

Hybrid Hyperchaotic Pseudorandom Number Generator with Dynamic Perturbation

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- Novel hybrid PRNG architecture based on a dynamic coupling scheme between existing 4D continuous and 2D discrete hyperchaotic systems.
- Dynamic parametric perturbation to mitigate digital signal degradation.
- Modular multiplication and non-linear processing enhance randomness.
- Efficient FPGA implementation using fixed-point arithmetic in VHDL.
- Validated via NIST 800-22 and TestU01 tests with a robust 224-bit key space.